Surge Harness Testing Report

Arva Syed and Jordyn Rabinowitz

Surge Harness is a wearable movement-resistance system designed to help climbers, athletes, and everyday users build strength and improve movement control safely. Instead of adding bulky external load, the system uses elastic resistance routed from the torso to the arms to gently slow motion, promote body awareness, and support mindful training and rehabilitation.

A collaboration between Arva Syed and Jordyn Rabinowitz, this is a project developed as part of the 2026 Senior Capstone for the Creative Technology and Design Bachelors of Science Program out of the ATLAS Institute at CU Boulder's School of Engineering.

Testing

As part of the development process for Surge Harness we conducted three rounds of focus group testing and a final climbing meetup that was open to the public.

We had a total of 7 participants consisting of three men and four women. There was a range of climbing experience with one beginner, one competitive climber, and five mid-level climbers. There was a range of shoulder mobility as well from a standard range of mobility to extremely hypermobile. There was also a range of history of shoulder injury from no shoulder injuries to multiple shoulder injuries.

Since these observations are distributed across so few participants we will be looking at distributions. More data is required for statistical significance.

Aquiring the Data

The data is mainly self-reported and testing participants were free to choose the difficulty of climbs attempted as well as the strength of resistance used with the harness. Participants were also asked to attempt the same climbs with and without the harness. We had a total of 7 participants each doing 8-11 climbs for a total of 59 climbs. Each climb will be analyzed as a discrete data point. Climbs were rated on difficulty on a scale of 1-11 to normalize across different gyms rating systems. All climbs were completed at or below a difficulty of 8.

Analyzing the Data

```r
library(tidyverse)
library(ggplot2)

#import climbing csv file
climbing <- read_csv("climbing.csv")

#wrangling
climbing$Resistance <- as.factor(climbing$Resistance)
#Filter out unsucessful climbs
climbingSuccess <- climbing %>%
  filter(Top == "y")
#Filter out harness vs no harness climbs
climbingHarness <- climbingSuccess %>%
```

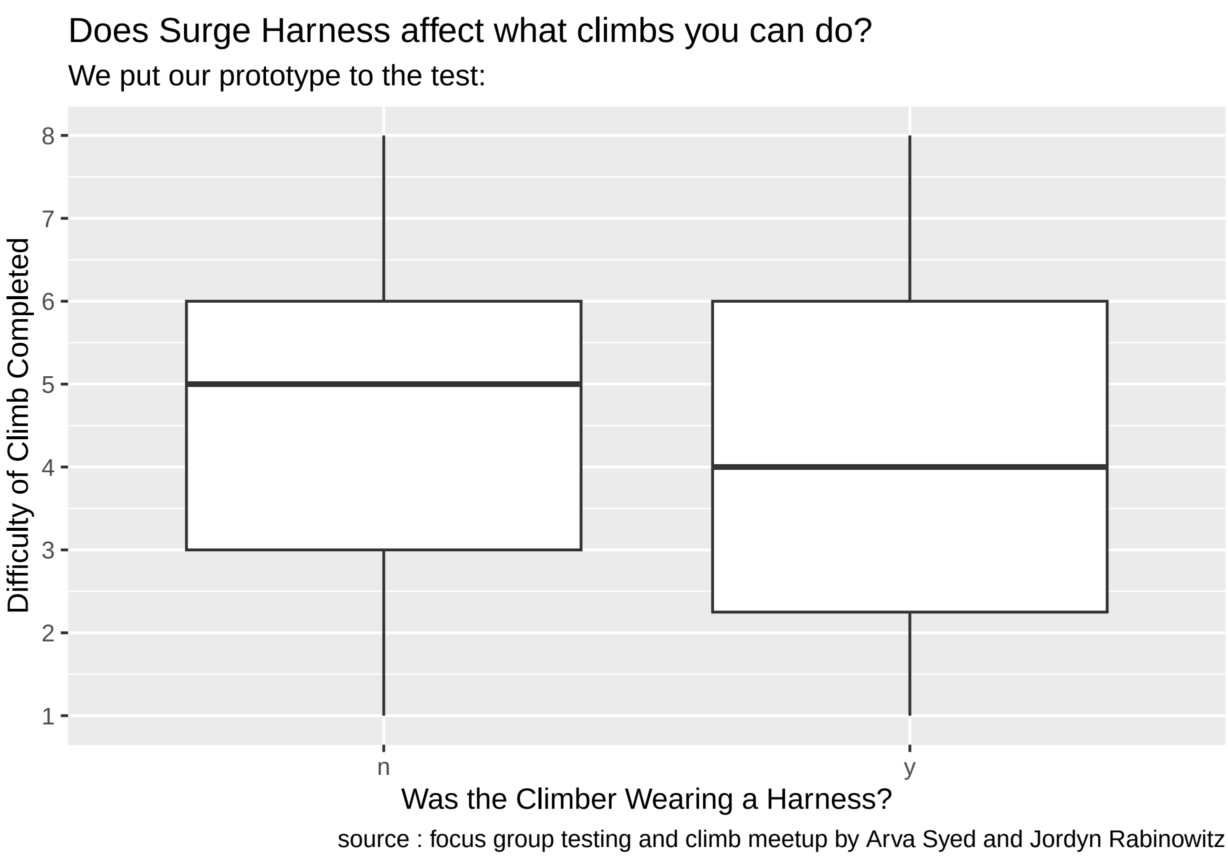
```r
  filter(Harness == "y")
climbingNoHarness <- climbingSuccess %>%
  filter(Harness == "n")

#With Harness
summary(climbingHarness$Points)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   1.000   2.250   4.000   4.167   6.000   8.000
```

```r
#Without Harness
summary(climbingNoHarness$Points)
```

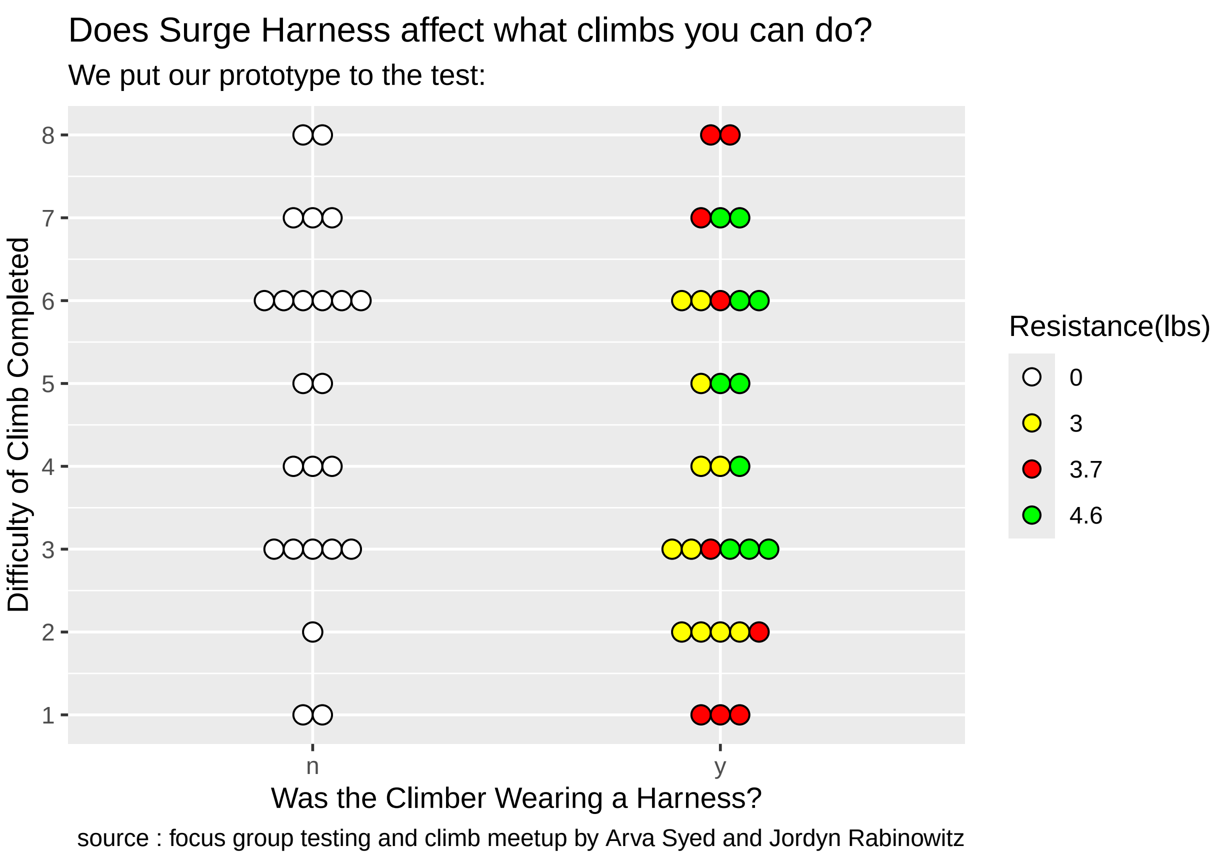
```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    1.00    3.00    5.00    4.75    6.00    8.00
```

```r
#visualize data distribution
ggplot(data=climbingSuccess, aes(x=Harness,y=Points))+
  geom_boxplot()+
  labs(title = "Does Surge Harness affect what climbs you can do?",
       subtitle = "We put our prototype to the test:",
       caption = "source : focus group testing and climb meetup by Arva Syed and Jordyn Rabinowitz",
       x = "Was the Climber Wearing a Harness?",
       y = "Difficulty of Climb Completed")+
  scale_y_continuous(limits=c(1,8),
                     n.breaks = 8)
```



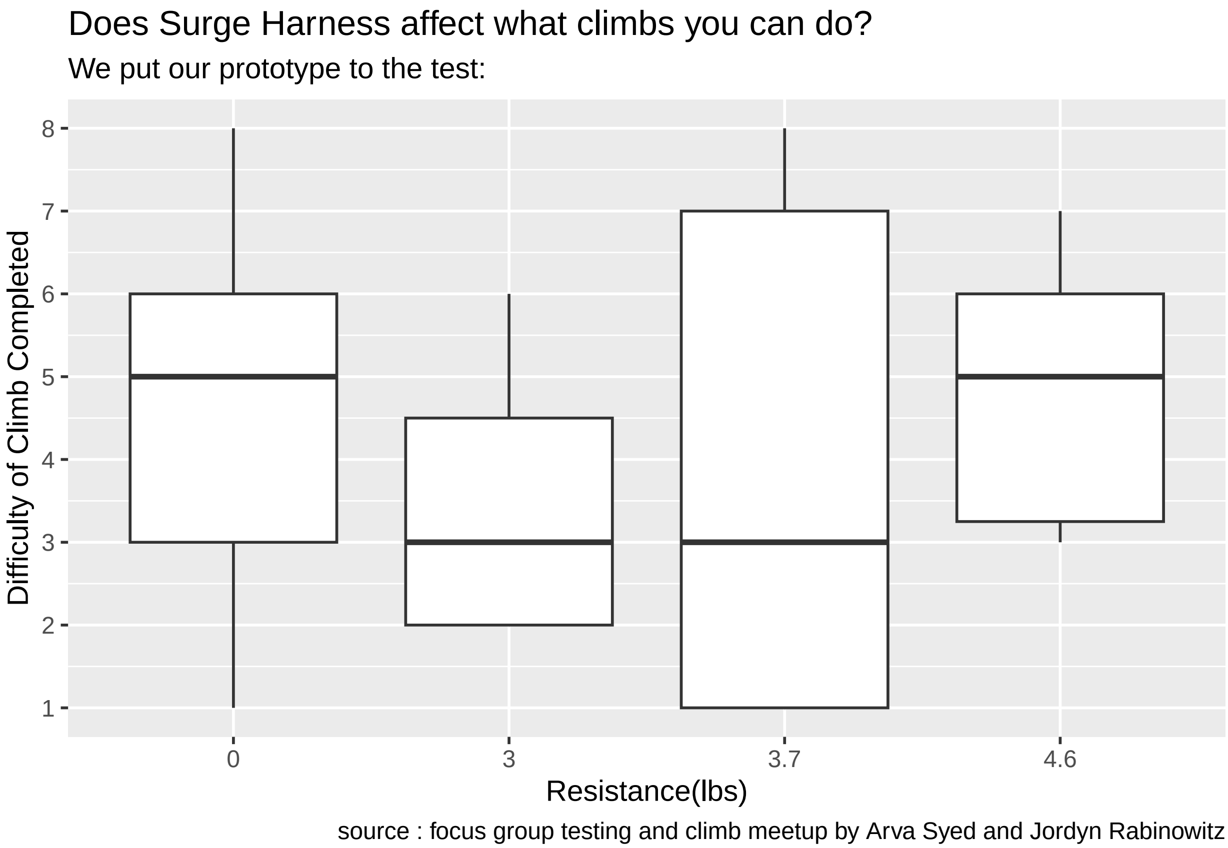

```r
ggplot(data=climbingSuccess, aes(x=Harness, y=Points))+
```

```r
  geom_dotplot(binaxis = "y",
               stackdir = "center",
               aes(fill = Resistance),
               stackgroups = T,
               binpositions="all")+
  labs(title = "Does Surge Harness affect what climbs you can do?",
       subtitle = "We put our prototype to the test:",
       caption = "source : focus group testing and climb meetup by Arva Syed and Jordyn Rabinowitz",
       x = "Was the Climber Wearing a Harness?",
       y= "Difficulty of Climb Completed",
       fill = "Resistance(lbs)")+
  scale_fill_manual(values = c("4.6" = "green",
                               "3.7" = "red",
                               "3" = "yellow",
                               "0" = "white")) +
  scale_y_continuous(limits=c(1,8),
                     n.breaks = 8)
```

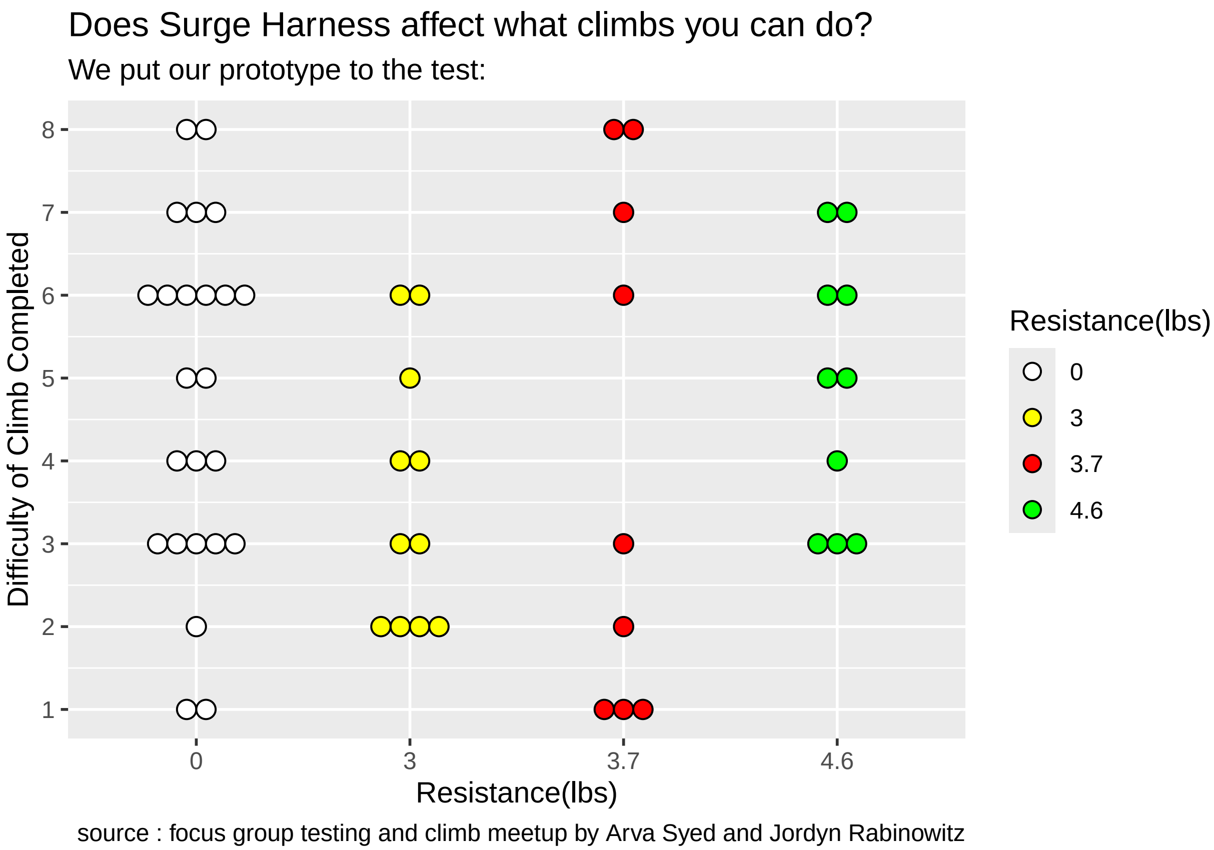



```r
#Break Down By Resistance
ggplot(data=climbingSuccess, aes(x=Resistance,y=Points))+
  geom_boxplot()+
  labs(title = "Does Surge Harness affect what climbs you can do?",
       subtitle = "We put our prototype to the test:",
       caption = "source : focus group testing and climb meetup by Arva Syed and Jordyn Rabinowitz",
       x = "Resistance(lbs)",
       y = "Difficulty of Climb Completed")+
  scale_y_continuous(limits=c(1,8),
```

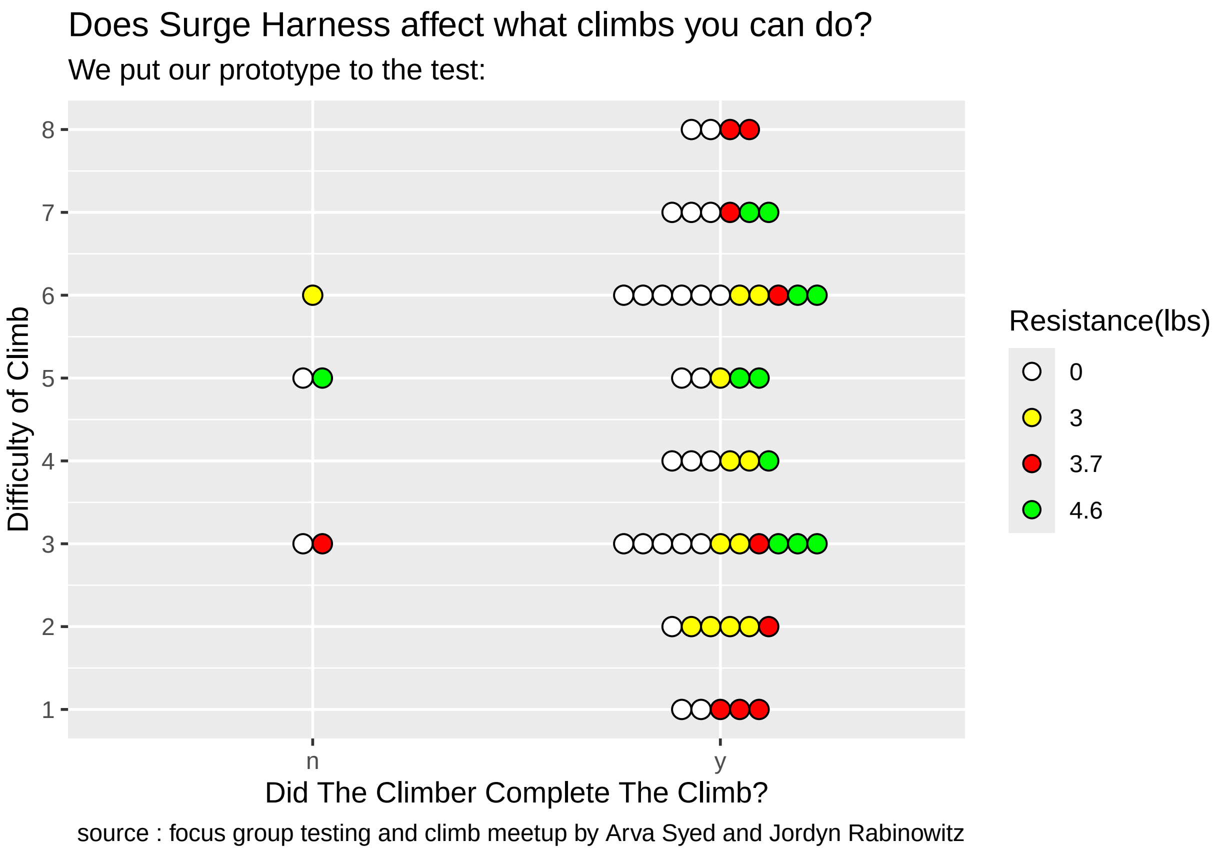
```
                     n.breaks = 8)
```



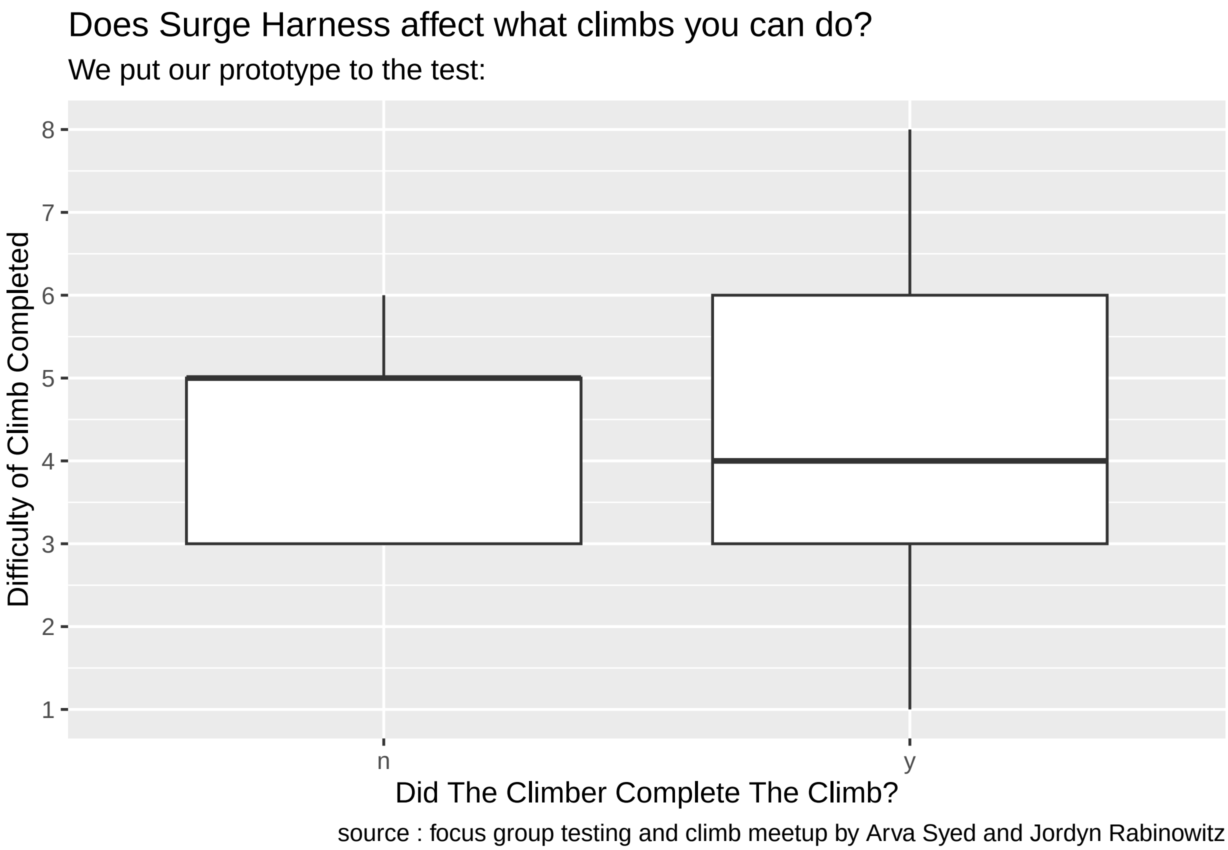

```
ggplot(data=climbingSuccess, aes(x=Resistance, y=Points))+
  geom_dotplot(binaxis = "y",
               stackdir = "center",
               aes(fill = Resistance),
               stackgroups = T,
               binpositions="all")+
  labs(title = "Does Surge Harness affect what climbs you can do?",
       subtitle = "We put our prototype to the test:",
       caption = "source : focus group testing and climb meetup by Arva Syed and Jordyn Rabinowitz",
       x = "Resistance(lbs)",
       y= "Difficulty of Climb Completed",
       fill = "Resistance(lbs)")+
  scale_fill_manual(values = c("4.6" = "green",
                               "3.7" = "red",
                               "3" = "yellow",
                               "0" = "white")) +
  scale_y_continuous(limits=c(1,8),
                     n.breaks = 8)
```




```r
#Break Down By Success or Failure
ggplot(data=climbing, aes(x=Top, y=Points))+
  geom_dotplot(binaxis = "y",
               stackdir = "center",
               aes(fill = Resistance),
               stackgroups = T,
               binpositions="all")+
  labs(title = "Does Surge Harness affect what climbs you can do?",
       subtitle = "We put our prototype to the test:",
       caption = "source : focus group testing and climb meetup by Arva Syed and Jordyn Rabinowitz",
       x = "Did The Climber Complete The Climb?",
       y= "Difficulty of Climb",
       fill = "Resistance(lbs)")+
  scale_fill_manual(values = c("4.6" = "green",
                               "3.7" = "red",
                               "3" = "yellow",
                               "0" = "white")) +
  scale_y_continuous(limits=c(1,8),
                     n.breaks = 8)
```




```
ggplot(data=climbing, aes(x=Top,y=Points))+
  geom_boxplot()+
  labs(title = "Does Surge Harness affect what climbs you can do?",
       subtitle = "We put our prototype to the test:",
       caption = "source : focus group testing and climb meetup by Arva Syed and Jordyn Rabinowitz",
       x = "Did The Climber Complete The Climb?",
       y = "Difficulty of Climb Completed")+
  scale_y_continuous(limits=c(1,8),
                     n.breaks = 8)
```



Conclusions

From these distributions the average completed climb with the harness was marginally less difficult than without. The difference is minimal and the overall distributions are comparable. There does not seem to be any significant negative affect to using the Surge Harness for climbing. The data set is fairly limited so more testing is encouraged to draw more confident conclusions.